View results

Respondent

57 Anonymous

64:48

Time to complete

A. General information about the respondant

1. * Country

Sierra Leone

2. * Name of organization(s) or entity(s) responsible for the preparation of the report

Sierra Leone Research & Education Network

3. * Website of organization(s) or entity(s) responsible for the preparation of the report

https://slren.edu.sl/

4. * Email address of organization(s) or entity(s) responsible for the preparation of the report

5. Phone number of organization(s) or entity(s) responsible for the preparation of the report

6. Please describe the role/mandate of your organization

Provision of fast internet access and advanced digital services to education and research community in Sierra Leone. Lead to development of the national open Science policy. Promote the development of infrastructure for open science

7. * Full name of officially designated contact person(s) who completed this survey

Dr Thomas Philip Songu

8. * Position of officially designated contact person(s) who completed this survey

CEO, SLREN and ICT Director, Njala University

9. * Email address of officially designated contact person(s) who completed this survey

10. Full Name(s) of designated official(s) certifying the report

The Minister, Technical & Higher Education, Sierra Leone. SG UNESCO Office, Sierra leone

11. Brief description of the consultation process established for the preparation of the report

- MTHE/SG nominated Dr Thomas Phlip Songu, CEO SLREN as National Focal Person for Open Science in Sierra Leone
- Communication received from UNESCO to acknowledge nomination
- Participated in a series of webinars organized by UNESCO on open science
- Open Science and STI ecosystem stakeholders were consulted about the 2024 country report and guidelines were shares
- Regular consultations made with stakeholders to obtain necessary information on STI and open science
- Inputs from stakeholders regarding open science were collated the for the preparation of this report

12. * Name of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

Ministry of Technical & Higher Education, National Science, Technology and Innovation Council

13. * Website of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey (Please use commas to separate multiple links, as: link A, link B)

https://www.mthe.gov.sl/

14. * Email address of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

15. Sector of other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey

- [x] Public
- [ ] Private
- [ ] Civil Society
- [x] Not for Profit making

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING

ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

Part 1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE,

ASSOCIATED BENEFITS AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

16. 1.1 Has the 2021 UNESCO Recommendation on Open Science been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

- (selected) Yes
- No

17. 1.1 cont. **If yes**, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

MTHE, MOCTI, NSATIC, SLREN, MDAs, Higher Education & Research Institutions (Universities), Civil Societies

18. 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

- (selected) Yes
- No

19. 1.2 cont. **If yes,** which language(s)?

English

20. 1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? *(ref.: (i) 16.f, 16.g, 16.h)*

[i] Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

- (selected) Yes
- No

21. 1.3 cont. **If yes**, please provide details on the activities organized or foreseen and links as relevant.

 National Open Science Symposium – August 2023 National Open Science valitation workshops – Feb 2024 Open science and STI sensitisation in universities and colleges.research institutions – December 2024

22. 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[i], in publicly funded research? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

[i] The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.

Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.

Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

- (x) Yes
- ( ) No

23. 1.4 cont. If **yes**, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

Part of the national open science policy implementation for 2025-2026

Part 2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

24. 2.1 Does your country have a national policy[i], strategy or plan of action on science, technology and innovation (STI)?

[i] National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

- (x) Yes
- ( ) No

25. 2.1 cont. **If yes**

· please share a machine-readable PDF file of the policy using this link: **https://tinyurl.com/27p5zct8** designating openscience@unesco.org as the recipient, and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*

In the comment box, please indicate the question number for this attachment / document.

After uploading, please return to this page to continue the survey.

2.1

26. 2.1 cont. **If yes:**
    - does it include any of the following open science elements:

- [x] open access to scientific knowledge
- [x] open science infrastructure
- [x] open engagement of societal actors
- [x] open dialogue with other knowledge systems

27. 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)[i] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

    [i] These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

- ○ Yes
- ● Partly
- ○ No

28. 2.2 cont. **If yes or partly,** please share a machine-readable PDF file(s) of the policy(ies) using this link: **https://tinyurl.com/27p5zct8** designating openscience@unesco.org as the recipient, and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

    copy sent

29. 2.3 In your country, are there specific policy instruments[i] that aim to promote open science?

    [i] Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

- ● Yes
- ○ Under development
- ○ No

30. 2.3 cont. **If yes or under development,** please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

    refer to copy of Open Science policy

31. 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? *(ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)*

- (•) Yes
- ( ) Under development
- ( ) No

32. 2.4 cont. **If yes or under development,** please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

refer to policy

33. 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? *(ref.: (ii) 17.d, 17.e, 17.g)*

- (•) Yes
- ( ) Under development
- ( ) No

34. 2.5 cont. **If yes,** please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).
To view the elements of open science, see: https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/379949eng.pdf.multi.nameddest=20/p.nameddest=20/.page=9

STI policy

35. 2.6 In your country, are there specific funding mechanisms[i] for open science at the national level? *(ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)*

[i] A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

- ( ) Yes
- ( ) Under development
- (•) No

36. 2.6 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

37. 2.7 1.1 In your country, have open science practices[i] been integrated in existing research funding mechanisms?

[i] Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

- ○ Yes
- ◉ No

38. 2.7 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

39. 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17.i)

- ○ Yes
- ◉ No

40. 2.8 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

41. 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.j, 23.c)

- ○ Yes
- ◉ Under development
- ○ No

42. 2.9 cont. **If yes or under development,** please provide details and links as relevant.

Part 3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

43. 3.1 What is the latest official data for your country’s percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

44. 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

45. 3.3 Are there national research and education networks (NRENs)[i] active in the country? *(ref.: (iii) 18.c)*

[i] A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

- (●) Yes
- ( ) No

46. 3.3 cont. **If yes**, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

https://slren.edu.sl/

47. 3.4 Are there national or institutional open science infrastructures[i] in your country? *(ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)*

[i] Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

- ( ) Yes
- (●) Under development
- ( ) No

48. 3.4 cont. **If yes or under development**, please indicate which type(s) of infrastructure:

- [ ] federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage
- [ ] community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- [x] platforms for exchanges and co-creation of knowledge between scientists and society
- [ ] community-based monitoring and information systems to complement national, regional and global data and information systems

49. 3.4 cont. **If yes or under development**, for each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

Deployment of DSPACE institutional repository

50. 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

- ○ Yes
- ● No

51. 3.5. cont. **If no**, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

- ☐ Yes
- ☐ No
- ☑ Preparing to connect to WACREN

52. 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.g)

- ● Yes
- ○ No

53. 3.6 cont. **If yes**, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open sciencei, stage of research, accessibility to all the relevant users in the region or internationally).

Member of the WACREN community

Part 4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY

AND CAPACITY BUILDING FOR OPEN SCIENCE

54. 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

- ● Yes
- ○ No

55. 4.1 cont. **If yes**, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

- National open science symposium - National open science policy valaidation workshop - Opon science sensitisation in universities and colleges - Technical training to develop open science infrastructure and network backbone for Sierra Leone

56. 4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

- ● Yes
- ○ No

57. 4.2 cont. **If yes**, please provide more information and links as relevant.

- National STI and Open Science sensitisation workshop for stakeholders - National STI and OS sensitisation in universitis and collegues

58. 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b)

- ○ Yes
- ◉ No

59. 4.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

60. 4.4 Are open educational resources[i] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? *(ref.: (iv) 19.d)*

[i] Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

- ◉ Yes
- ○ Partly
- ○ No

61. 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

- ◉ Yes
- ○ No

62. 4.5 cont. **If yes**, please provide more information and links as relevant.

- National STI and Open Science sensitisation workshop for stakeholders - National STI and OS sensitisation in universitis and collegues

Part 5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

63. 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? *(ref.: (v) 20.b, 20c, 20i)*

- (●) Yes
- ( ) No

64. 5.1 cont. **If yes**, please provide more information and links as relevant.

Refer to the national policy

65. 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c, 20.d)

- (●) Yes
- ( ) No

66. 5.2 cont. **If yes**, please provide details and links as relevant.

ODL policy, OER policies

67. 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)

- ( ) Yes
- (●) No

68. 5.3 cont. **If no**, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

69. 5.4 Have there been any unintended negative consequences[i] of open science practices reported in your country? (ref.: (v) 20)

[i] For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

- ( ) Yes
- (●) No

Part 6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES

OF THE SCIENTIFIC PROCESS

70. 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? *(ref.: (vi) 21)*

- ○ YES, at the national level
- ○ YES, at the institutional level
- ◉ YES, at both the national and institutional levels
- ○ No

71. 6.1 cont. **If yes**, please indicate which of the recommended actions they address:

- [x] promotion of pre-prints
- [x] exploration of open peer-review practices
- [x] valuing and sharing of negative scientific results and associated data
- [x] participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- [x] development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas
- [x] promoting open innovation practices
- [ ] Other

72. 6.1. cont. **If yes**, please provide more information and links as relevant

refer to policy

Part 7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT

OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

73. 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Partnership with WACREN, Other Universities, SGCIs

74. 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e, 22.f)

- ◉ Yes
- ○ No

75. 7.2 cont. **If yes**, please provide links as relevant and more information.

76. 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: (vii) 22.c)

- (●) Yes
- ( ) No

77. 7.3 cont. **If yes**, please provide links as relevant and more information.

Part 8. GENERAL CONSIDERATIONS

78. 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

- ( ) Yes
- (●) No

79. ANY OTHER COMMENTS: If you wish, you may enter any additional comments or information here. You may also contact openscience@unesco.org